

SUMMER INTERNSHIP PROJECT
INSPIRE MENTORSHIP GRANT

ON

**STUDYING THE BASICS OF CIRCUIT-BASED QUANTUM
COMPUTING**

Submitted by

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Last but not least, I express my heartfelt thanks to my family and friends for their constant support and kindness. Their unwavering encouragement and belief in my abilities have been a source of strength and motivation throughout this project. I am deeply grateful for their understanding and patience during the challenging times, and for always being there to celebrate the successes and provide comfort during setbacks. Their love and support have played a crucial role in my achieving my goals.

Studying the Basics of Circuit-Based Quantum Computing

Objectives of the Project

1. Study the fundamentals of quantum computing with a focus on circuit-based models.
2. Develop hands-on experience with the PennyLane Python library, a cross-platform tool for quantum computing, quantum machine learning, and quantum chemistry, to enhance understanding of quantum computing concepts.
3. Implement and explore the algorithms learned using PennyLane.

Introduction of Research Topic

Quantum computing is a rapidly emerging multidisciplinary field that integrates computer science, physics, and mathematics, using principles of quantum mechanics like superposition and quantum interference to tackle certain problems faster than their classical counterparts.

Potential applications of quantum computing include machine learning (ML), cryptography, optimization, and simulating physical systems. Future use cases may involve portfolio optimization in finance or simulating chemical systems, addressing challenges currently beyond the reach of the most advanced supercomputers.

One of the most prominent models of quantum computing is circuit-based quantum computing. In this model, quantum algorithms are expressed as quantum circuits, which are composed of a sequence of quantum gates acting on qubits. These quantum gates are the building blocks of quantum circuits and perform operations that change the state of qubits based on the principles of quantum mechanics.

PennyLane is a versatile Python library designed for programming quantum computers across multiple platforms. Its differentiable programming model allows for the execution and training of quantum programs on a variety of hardware backends. It is a cutting-edge software framework designed for quantum machine learning, quantum computing, and quantum chemistry. Developed by Xanadu, PennyLane bridges the gap between quantum computing and classical machine learning, allowing researchers and developers to harness the power of quantum hardware alongside classical machine learning libraries.

In this project, we explore various algorithms and implement them on a classical computer using PennyLane software.

Theoretical Framework of Topic

Name	No. of qubits	Operator symbol	Matrix
Identity	1	I	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
Pauli X, NOT, bit flip	1	X, NOT, σ_x	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli Y	1	Y, σ_y	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli Z, phase flip	1	Z, σ_z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard	1	H	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Controlled-NOT	2	CNOT, XOR	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

Table 1: Table of Quantum Gates

The following algorithms and concepts were worked upon:

1 Deutsch's problem

Let's consider functions that map one qubit to another. There are four such functions:

Function	$x=0$	$x=1$
f_0	0	0
f_1	0	1
f_2	1	0
f_3	1	1

Table 2: Functions mapping one qubit to another

We want to determine whether f is constant (i.e., $f(0) = f(1)$), satisfied by f_0 and f_3 , or not constant

(i.e., $f(0) \neq f(1)$), satisfied by f_1 and f_2 .

Solution

We start with both input and output registers in the state $|0\rangle$, and apply the following operators:

$$(H \otimes H)(X \otimes X)|0\rangle|0\rangle = \frac{1}{2}(|0\rangle \otimes |0\rangle - |1\rangle \otimes |0\rangle - |0\rangle \otimes |1\rangle + |1\rangle \otimes |1\rangle).$$

.

If we take this state as input to U_f , then the resulting state is

$$\frac{1}{2}[U_f(|0\rangle|0\rangle) - U_f(|1\rangle|0\rangle) - U_f(|0\rangle|1\rangle) + U_f(|1\rangle|1\rangle)].$$

The action of U_f is simply

$$\frac{1}{2}[|0\rangle|f(0)\rangle - |1\rangle|f(1)\rangle - |0\rangle|\tilde{f}(0)\rangle + |1\rangle|\tilde{f}(1)\rangle],$$

So if $f(0) = f(1)$, the output state is

$$\frac{1}{2}[|0\rangle - |1\rangle]|f(0)\rangle - |\tilde{f}(0)\rangle], \quad f(0) = f(1).$$

But if $f(0) \neq f(1)$, the output state becomes

$$\frac{1}{2}[|0\rangle + |1\rangle]|f(0)\rangle - |\tilde{f}(0)\rangle], \quad f(0) \neq f(1).$$

Finally, if we apply a Hadamard transformation to the input register, these become

$$|1\rangle \otimes \frac{1}{\sqrt{2}}(|f(0)\rangle - |\tilde{f}(0)\rangle), \quad f(0) = f(1),$$

$$|0\rangle \otimes \frac{1}{\sqrt{2}}(|f(0)\rangle - |\tilde{f}(0)\rangle), \quad f(0) \neq f(1).$$

Hence we have,

$$H \otimes I U_f H \otimes H X \otimes X |0\rangle|0\rangle = \begin{cases} |1\rangle \otimes \frac{1}{\sqrt{2}}(|f(0)\rangle - |\tilde{f}(0)\rangle), & \text{if } f(0) = f(1), \\ |0\rangle \otimes \frac{1}{\sqrt{2}}(|f(0)\rangle - |\tilde{f}(0)\rangle), & \text{if } f(0) \neq f(1). \end{cases}$$

Therefore the state of the input register is determined to be $|1\rangle$ or $|0\rangle$ depending on whether or not $f(0) = f(1)$, so by measuring the input register we can indeed answer the question of whether $f(0) = f(1)$ or not.

2 The Bernstein–Vazirani Problem

Let a be an unknown non-negative n -bit integer. We are given a function $f(x)$ defined as:

$$f(x) = a \cdot x = a_0 x_0 \oplus a_1 x_1 \oplus \dots \oplus a_{n-1} x_{n-1}$$

Our goal is to determine the integer a .

Solution

We use an n -qubit input register and a 1-qubit output register. The initial state of the output register is $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Applying the function U_f to $|x\rangle|y\rangle$ flips the output qubit if $f(x) = 1$, transforming the state to:

$$U_f|x\rangle|y\rangle = \frac{1}{\sqrt{2}}|x\rangle \left(|0\rangle - (-1)^{f(x)}|1\rangle \right)$$

After applying U_f , the state is:

$$\frac{1}{\sqrt{2}}|x\rangle \left(|0\rangle - (-1)^{f(x)}|1\rangle \right)$$

Next, applying $H^{\otimes n}$ to $|x\rangle$:

$$H^{\otimes n}|x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{x \cdot y} |y\rangle$$

Applying the unitary transformations:

$$(H^{\otimes n} \otimes I)(U_f)(H^{\otimes n} \otimes H)|0^n\rangle|1\rangle = \frac{1}{2^n} \sum_{x=0}^{2^n-1} \sum_{y=0}^{2^n-1} (-1)^{f(x)+x \cdot y} |y\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

The sum:

$$\sum_{x=0}^{2^n-1} (-1)^{(a \cdot x) + (y \cdot x)}$$

is zero unless $y = a$, leading to:

$$(H^{\otimes(n+1)})(U_f)(H^{\otimes(n+1)})|0^n\rangle|1\rangle = |a\rangle|1\rangle$$

Thus, measuring the input register reveals a .

3 Toffoli Gates

The Toffoli gate, also known as the CCNOT gate (Controlled-Controlled-NOT gate), is a fundamental quantum gate used in quantum computing. It is a three-qubit gate that performs a conditional operation based on the states of two control qubits.

4 RSA Encryption and Period Finding

4.1 RSA Encryption

RSA is a public-key cryptosystem based on the difficulty of factoring large primes. The steps are:

1. **Key Generation:** Choose two large primes p and q . Compute $n = p \times q$ and $(p-1) \times (q-1)$. Select a public exponent c coprime to $(p-1) \times (q-1)$, and find d such that $d \times c \equiv 1 \pmod{(p-1) \times (q-1)}$. The public key is (n, c) and the private key is (n, d) .
2. **Encryption:** Convert the message m to an integer, then compute $b \equiv m^c \pmod{n}$ using the public key.
3. **Decryption:** Recover the message m by computing $m \equiv b^d \pmod{n}$ with the private key.

RSA's security depends on the difficulty of factoring n .

4.2 Period Finding and RSA Encryption

Period finding algorithms, such as Shor's algorithm, are crucial for efficient integer factorization. They find the period of a function related to modular exponentiation, which aids in factoring the modulus n in RSA encryption. This ability to quickly factor large numbers poses a significant threat to RSA encryption, as it can compromise the private key.

5 Quantum Fourier Transformation

The n -qubit Fourier Transform U_{FT} is defined as:

$$U_{FT}|x_n\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} e^{2\pi i xy/2^n} |y_n\rangle$$

This can be expressed in terms of Hadamard gates and phase gates as follows:

Hadamard Transform:

$$H^{\otimes n}|x_n\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} e^{\pi i x \cdot y} |y_n\rangle$$

Phase Gates Z_x :

$$Z|y_n\rangle = e^{\frac{2\pi i y}{2^n}} |y_n\rangle$$

Thus:

$$U_{FT}|x_n\rangle = Z^x H^{\otimes n}|0_n\rangle$$

Rewriting Z_x in terms of phase gates:

$$Z_x = \exp\left(i\pi\left(x_0 n_3 + \left(x_1 + \frac{1}{2}x_0\right)n_2 + \left(x_2 + \frac{1}{2}x_1 + \frac{1}{4}x_0\right)n_1 + \left(x_3 + \frac{1}{2}x_2 + \frac{1}{4}x_1 + \frac{1}{8}x_0\right)n_0\right)\right)$$

Combining Results:

$$U_{FT}|x_3x_2x_1x_0\rangle = H_3 \cdot V_{32}H_2 \cdot V_{31}V_{21}H_1 \cdot V_{30}V_{20}V_{10}H_0|x_0x_1x_2x_3\rangle$$

Where:

$$V_{ij} = \exp\left(\frac{i\pi n_i n_j}{2|i-j|}\right)$$

Final Form:

$$U_{FT} = H_3(V_{32}H_2)(V_{31}V_{21}H_1)(V_{30}V_{20}V_{10}H_0)P$$

Here, P represents the permutation that reverses the qubit order.

6 Grover's Search Algorithm

Grover's algorithm utilizes a quantum search subroutine to identify a special n -bit integer a . This subroutine returns a function $f(x)$ such that:

$$\begin{aligned} 0 & \text{ if } x \neq a \\ 1 & \text{ if } x = a \end{aligned}$$

Classically, finding a would involve checking each candidate until $f(x) = 1$, which takes (2^n) steps. Grover's algorithm significantly improves this with a quantum computational speed-up, using a unitary transformation U_f . This transformation operates on an n -qubit input register (containing x) and a 1-qubit output register, flipping the output qubit if x is the special number a :

$$U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$$

The 1 qubit output register is prepared in the state:

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Action of U_f :

$$U_f(|x\rangle \otimes H|1\rangle) = (-1)^{f(x)}(|x\rangle \otimes H|1\rangle)$$

The effect of U_f on $|x\rangle \otimes H|1\rangle$ is the same as doing nothing to the 1-qubit output register while applying an n -qubit unitary transformation V on the input register:

$$V|x\rangle = (-1)^{f(x)}|x\rangle = \begin{cases} |x\rangle, & x \neq a, \\ -|a\rangle, & x = a. \end{cases}$$

Since U_f is linear, so is V . Acting on a superposition $|\psi\rangle = \sum_x |x\rangle\langle x|\psi\rangle$, V changes the sign of the component along $|a\rangle$ while leaving the orthogonal component unchanged:

$$V|\psi\rangle = |\psi\rangle - 2|a\rangle\langle a|\psi\rangle.$$

Let $V = 1 - 2|a\rangle\langle a|$, where $|a\rangle\langle a|$ is the projection operator on state $|a\rangle$. In Grover's algorithm, U_f is

the only unitary transformation that affects the output register, which remains unentangled in the state $H|1\rangle$. Therefore, we can simplify:

$$U_f(|\psi\rangle \otimes H|1\rangle) = (|\psi\rangle - 2|a\rangle\langle a|\psi\rangle) \otimes H|1\rangle$$

We replace the $(n+1)$ -qubit U_f with the n -qubit V acting on the input register and define other operators to act only on the input register.

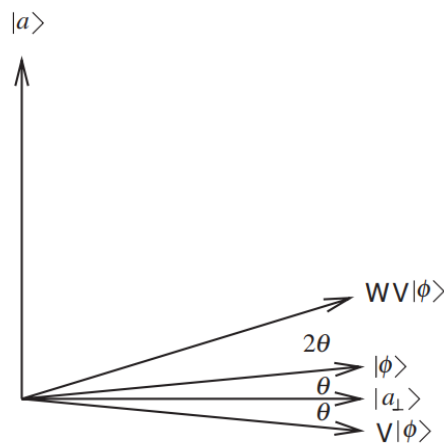
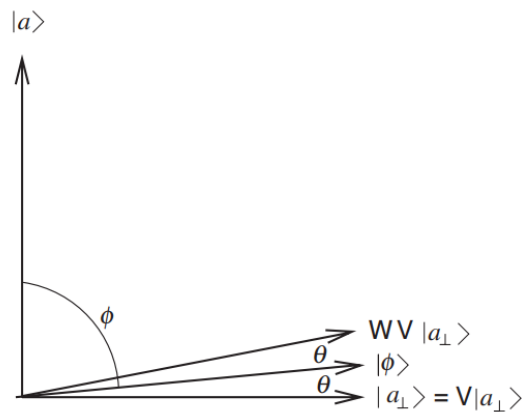
o execute Grover's algorithm, transform the n -qubit input register into the uniform superposition of all possible inputs:

$$|\phi\rangle = H^{\otimes n}|0^n\rangle = \frac{1}{2^{n/2}} \sum_{x=0}^{2^n-1} |x\rangle.$$

In addition to V , Grover's algorithm requires a second n -qubit unitary W that acts on the input register, preserving the component along $|\phi\rangle$ while changing the sign of its orthogonal component:

$$W = 2|\phi\rangle\langle\phi| - 1,$$

where $|\phi\rangle\langle\phi|$ is the projection operator on $|\phi\rangle$. Grover's algorithm involves repeatedly applying the product WV to the input register, initially in the state $|\phi\rangle$.



Hence, we can see that WV is a rotation. Because WV is a rotation, applying it to any vector in the plane rotates that vector through the angle 2θ from $|a_\perp\rangle$ to $|\phi\rangle$. Thus, applying WV to the initial state $|\phi\rangle$ gives a vector rotated by 3θ , since $|\phi\rangle$ is already rotated by θ from $|a^\perp\rangle$. Each subsequent application of WV increases the angle between the final state and $|a_\perp\rangle$ by 2θ .

Since θ is very close to $2^{-n/2}$, after approximately

$$\frac{\pi}{4} \cdot 2^{n/2}$$

applications, the resulting state will be nearly orthogonal to $|a^\perp\rangle$ in the plane spanned by $|\phi\rangle$ and $|a\rangle$, and will be very close to $|a\rangle$.

7 Quantum Error Correction

Quantum error correction is a field of quantum computing focused on protecting quantum information from errors due to decoherence and other quantum noise. Unlike classical error correction, quantum error correction must handle the unique properties of quantum information, such as superposition and entanglement. We learn to correct bit-flip errors, phase errors and also combinations of both.

8 Quantum Cryptography

Quantum cryptography leverages the principles of quantum mechanics to enhance the security of communication. Unlike classical cryptographic methods, which rely on mathematical algorithms and computational complexity, quantum cryptography uses the fundamental properties of quantum mechanics to provide security guarantees. The BB84 protocol, developed by Charles Bennett and Gilles Brassard in 1984, is one of the first and most widely studied quantum key distribution (QKD) protocols. It uses the principles of quantum mechanics to enable two parties to securely share a cryptographic key.

Profile of Organistaion/Research Lab

The Indian Institute of Science Education and Research (IISER) Pune, established in 2006 by the Ministry of Education , is a leading institution focused on research and education in the basic sciences. Since then. IISER Pune has committed towards the advancement of education, research, and community service to benefit society. They strive to foster an environment rooted in integrity, fairness, dignity, and professionalism, ensuring equal opportunities for everyone. In 2012, it was recognized as an Institute of National Importance by an Act of Parliament.

The Department of Physics has active research groups in the areas such as Astrophysics and Cosmology; Atomic, Molecular, Optical and Plasma Physics; Condensed Matter Physics; Gravitational Physics; High Energy Physics; Non-linear Dynamics and Statistical Physics and Soft and Living Matter Physics.

Dr. Sreejith G J joined IISER Pune in January 2017 in Deartment of Physics. He is a condensed matter theorist specializing in strongly correlated quantum many-body systems and statistical mechanics. His research focuses on the fractional quantum Hall effect and simple spin systems. He primarily employs computational and semi-analytical methods and has a keen interest in developing algorithms to facilitate the study of interacting many-body systems.

Methodology Followed

1. **Daily Reading:** Each day, a portion of the reference book was read and understood to build a strong theoretical foundation.
2. **Daily Meetings:** Had daily meetings with the guide to discuss the day's readings, clarify doubts, and gain deeper insights.
3. **Supplement with Practical Tutorials:** Used Pennylane's tutorials and documentation to get hands-on experience with its functionalities and usage.
4. **Apply Theoretical Knowledge:** Wrote and executed codes in Pennylane, starting with simple examples and gradually progressing to more complex implementations.
5. **Testing and Debugging:** Regularly tested and debugged the code to ensure correctness and efficiency of the implementations.
6. **Collaboration and Guidance:** Collaborated with peers or sought guidance from the guide to enhance the learning experience and gain diverse perspectives on problem-solving approaches.
7. **Code Compilation:** All code was written and compiled in Jupyter Notebook to facilitate interactive development and testing.

Analysis and Interpretation of Project

Quantum computing leverages quantum mechanics to perform computations beyond classical capabilities, allowing certain problems to be solved significantly faster. For instance, Deutsch's problem can be solved with a single quantum evaluation, while classical methods require two, thanks to quantum parallelism. Similarly, Bernstein-Vazirani's problem requires n classical evaluations but only one quantum evaluation.

Toffoli gates are vital in quantum computing for reversible computation, preserving information and reducing energy dissipation. They are essential for building universal quantum circuits and implementing quantum error correction.

The quantum period-finding algorithm poses a threat to RSA encryption by simplifying the factoring problem using quantum principles and the Quantum Fourier Transform, potentially breaking the encryption.

In the unstructured search problem, also known as black-box problem, classical methods are often less efficient than random testing. However, Grover's algorithm provides a $\frac{1}{\sqrt{N}}$ speed-up over random testing.

Quantum error correction is crucial for feasible quantum computation but challenging, as measuring qubits directly collapses the quantum state. Researchers have developed practical solutions to address this complex issue.

Quantum cryptography employs quantum mechanics to securely distribute keys, making eavesdropping detectable.

All of these examples demonstrate that quantum computing can significantly speed up various problems and enhance computational efficiency across multiple fields.

Implementing these algorithms face several significant challenges. Current quantum hardware limitations, such as restricted qubit counts impact the practical execution of these algorithms. Additionally, issues related to quantum decoherence and noise complicate the maintenance of quantum states and accuracy of computations. Effective quantum error correction is crucial but remains resource-intensive and complex to implement. As a result, while these quantum algorithms offer theoretical advantages, practical applications are constrained by the limitations of existing quantum technologies.

Conclusion and Suggestion of Project

Quantum computing helps in solving the present computationally hard tasks with much more speed and efficiency than the existing classical computers using the principles of quantum mechanics. By performing many parallel calculations simultaneously, quantum computers can significantly reduce computational time.

For instance, in cryptography, a quantum computer can solve integer factorization much faster than can classical methods by means of Shor's algorithm which might be an end-run for traditional encryption like RSA. At the same time, Grover's algorithm presents a quadratic speed-up in solving unstructured search problems by decreasing the time needed compared to traditional search methods. This almost unbelievable acceleration that the quantum computers allows can lead to advances in various areas like cryptography, optimization, and scientific research, and thus, it can bring about revolutionary changes that technology and industry will undergo.

Quantum computing research is actively carried out, which influences the development of various industries. Some of the most distinctive projects are carried by IBM, Google, Microsoft, Rigetti etc.

Despite significant progress, quantum computing still faces several technological challenges that limit its current applications, scalability, and reliability.

Quantum systems are delicate and susceptible to errors due to qubit fragility, interconnection issues, decoherence, and external noise. To address this, we need to enhance fault-tolerant structures, improve qubit coherence times, and develop better error correction methods.

Additionally, creating quantum hardware—such as cryogenic cooling systems, control circuits etc. remains a complex task that requires collaboration across various fields and innovative approaches.

To advance quantum computing, several key strategies should be pursued. Enhancing quantum hardware is crucial, including improving qubit quality, developing advanced cryogenic systems, and scaling up quantum systems. Robust error correction methods need to be developed and optimized to protect quantum information and reduce resource overhead. Expanding quantum algorithms and software will help solve a broader range of problems and make quantum computing more accessible. Interdisciplinary collaboration and industry-academia partnerships are essential for addressing the complex challenges of quantum technology. Additionally, advancing quantum communication and increasing public awareness and education about quantum computing will support the field's growth and application.

Results Achieved

A comprehensive understanding of various theoretical concepts was developed, which was then applied through implementations using the PennyLane framework. This approach allowed for practical exploration and testing of these concepts in a quantum computing environment.

Here are some examples of implementations:

```
DEUTSCH'S ALGORITHM

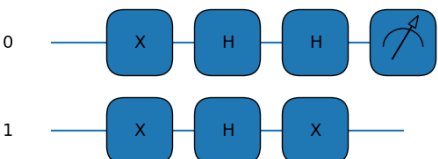
import pennylane as qml
import numpy as np

# Let the function be f_3 (constant function). It gives 1 for both input 0 and 1.

def deutsch1():
    qml.PauliX(wires=0)
    qml.PauliX(wires=1)
    qml.Hadamard(wires=0)
    qml.Hadamard(wires=1)
    qml.PauliX(wires=1)
    qml.Hadamard(wires=0)
    return qml.expval(qml.PauliZ(0))
dev1=qml.device('default.qubit',wires=2,shots=1000)
circuit1=qml.QNode(deutsch1,dev1)

result=circuit1()
print(result)
if result==1:
    print("The function is constant")
qml.draw_mpl(circuit1())

-1.0
The function is constant
[2]: (<Figure size 600x300 with 1 Axes>, <Axes: >)
```



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```
BERNSTEIN-VAZIRANI PROBLEM

import pennylane as qml
import numpy as np

def uf():
    qml.CNOT(wires=[0,4])
    qml.CNOT(wires=[2,4])

def bernstein():
    qml.PauliX(wires=4)

    for i in range(5):
        qml.Hadamard(wires=i)

    uf()

    for i in range(4):
        qml.Hadamard(wires=i)

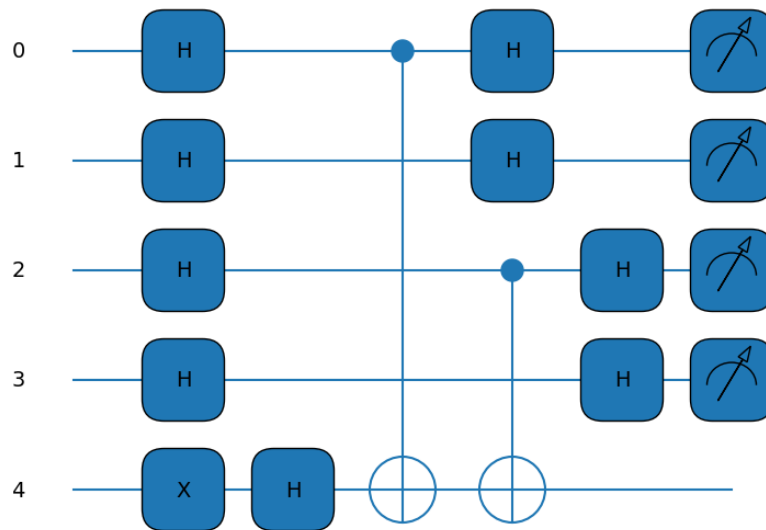
    return qml.sample(wires=range(4))

dev=qml.device('default.qubit',wires=5,shots=1)

circuit=qml.QNode(bernstein,dev)

print(circuit())

[1 0 1 0]
```



TOFFOLI GATES

```
10]: import pennylane as qml
import numpy as np

import numpy as np

sqrtX = 1/np.sqrt(2)*( np.array([[1, -1], [1, 1]]))
sqrtXd = np.conj(sqrtX).T

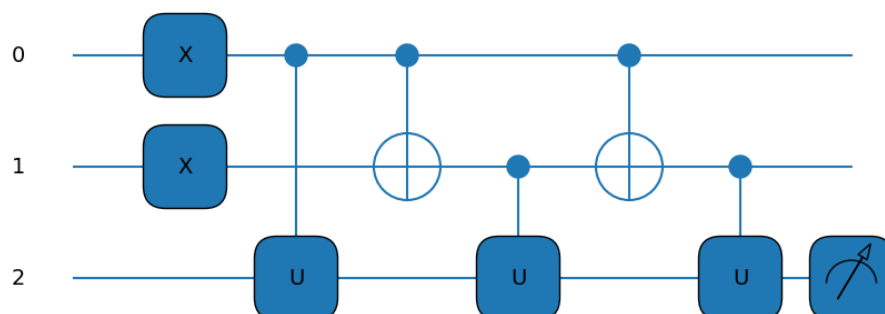
sqrtXd=np.conj(sqrtX).T
def quantumfn():
    qml.PauliX(wires=1)
    qml.PauliX(wires=0)
    qml.ControlledQubitUnitary(sqrtX,control_wires=0,wires=2)
    qml.CNOT(wires=[0, 1])
    qml.ControlledQubitUnitary(sqrtXd,control_wires=1,wires=2)
    qml.CNOT(wires=[0, 1])
    qml.ControlledQubitUnitary(sqrtX,control_wires=1,wires=2)
    return qml.expval(qml.PauliZ(2))

dev=qml.device('default.qubit',wires=3,shots=1000)

circuit=qml.QNode(quantumfn,dev)

print(circuit())

-1.0
```



```
[ ]: FACTORING

[42]: def function(b,x,n):
      return pow(b,x,n)

      b=2
      n=10117
      result1 = function(b,1,n)
      result2 = function(b,2,n)

      for x in range(3,10117):
          result3=(pow(b,x,n))
          if result3==result1:
              print(x-1)
              break
          elif result3==result2:
              print(x-2)
              break

330

[43]: z=pow(2,165,10117)
      print(z-1)
      print(z+1)

8909
8911

[44]: import math

      # Function to find GCD of two numbers
      def find_gcd(a, b):
          return math.gcd(a, b)

      # Example usage:
      num1 = 8909
      num2 = 10117

      gcd_result = find_gcd(num1, num2)
      print(f"The GCD of {num1} and {num2} is: {gcd_result}")

The GCD of 8909 and 10117 is: 151
```

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Quantum error correction code- 3 Qubit

```
[7]: import pennylane as qml
      import numpy as np

      dev=qml.device('default.qubit',wires=5,shots=1000)

      def errorcorrection():
          qml.PauliX(wires=2)
          qml.PauliX(wires=0)
          qml.CNOT(wires=[2,4])
          qml.CNOT(wires=[1,4])
          qml.CNOT(wires=[1,3])
          qml.CNOT(wires=[0,3])
          qml.Toffoli(wires=[3,4,1])
          qml.PauliX(wires=4)
          qml.Toffoli(wires=[3,4,0])
          qml.PauliX(wires=4)
          qml.PauliX(wires=3)
          qml.Toffoli(wires=[3,4,2])

          return [qml.expval(qml.PauliZ(i)) for i in range(3)]

      circuit=qml.QNode(errorcorrection,dev)
      result=circuit()
      print(result)

      qml.draw_mpl(circuit)()

[-1. -1. -1.]

[7]: (<Figure size 1300x600 with 1 Axes>, <Axes: >)
```


References

1. Mermin ND. Quantum Computer Science: An Introduction. Cambridge University Press; 2007.
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DECLARATION

I, **Athira Chandrasekharan**, hereby declare that the details/facts mentioned above are true to the best of my knowledge and I shall solely be held responsible in case of any discrepancies found in the details mentioned above.

Signature of Scholar

Date:

Place: